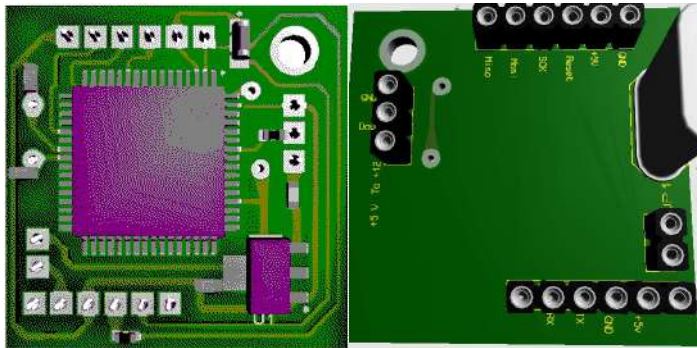


RANGMANG

MODEL	VOLTAGE	TYPE	ADDRESSABLE	NOTES
WS2811	12V	DIGITAL	YES	SUITABLE FOR LONG STRIPS
WS2812B	5V	DIGITAL	YES	BUILT-IN CHIP IN EACH LED
WS2813	5V	DIGITAL	YES	RESISTANT TO DATA LINE FAILURE
WS2815	12V	DIGITAL	YES	INDUSTRIAL-GRADE, LOW VOLTAGE DROP



RANGMANG Dance Mode Configuration

This configuration defines how the RANGMANG module controls LED dance modes. Each row encodes the following parameters (ADRGBSEM format):

A (Address) – 2 digits (A01–A98 for module, A99 for all).

D (Mode) – D01–D38 for dance styles, D99 for random dance selection.

R / G / B – 3 digits each. Normal range R000–R250, G000–G250, B000–B250.

Special values:

R251–R255 → Random Red

G251–G255 → Random Green

B251–B255 → Random Blue

S (Dance Time 1) – 3 digits, milliseconds per pixel move (first timing), range S000–S999.

E (Dance Time 2) – 3 digits, milliseconds per pixel move (second timing), range E000–E999.

M (Pixel Count) – 3 digits, number of dancing pixels. - ATmega328: M001–M480 - ATmega128: M001–M999

A (Channel)	D (Mode)	R (Red)	G (Green)	B (Blue)	S (Dance Time
1)	E (Dance Time 2)	M (Pixel Count)			
A01	D12	R251	G251	B251	S001 E001 M128

RANGMANG Fixed Color Mode Configuration (ATmega128 – 999 Pixels)

A (Address) – 2 digits (A01–A98 for module, A99 for all).

D (Mode) – must always be D00 for solid color.

R / G / B – 3 digits each, range R000–R255, G000–G255, B000–B255.

S (Start Pixel) – 3 digits, range S000–S999.

E (End Pixel) – 3 digits, range E000–E999.

M (Memory Slot) – 3 digits, M000 without save, M001–M999 for slots.

A (Channel)	D (Mode)	R (Red)	G (Green)	B (Blue)	S (Start Pixel)
	E (End Pixel)	M (Memory Slot)			

A01	D00	R200	G050	B100	S001	E120	M010
-----	-----	------	------	------	------	------	------

Power Regulation Notes

For 5V LED strips: You can remove the regulator and directly power the microcontroller from the strip's 5V socket.

For 12V LED strips: You must use a 12V-to-5V regulator to power the microcontroller safely.

Note: To prevent reverse connection of the LED strip to the module, use a properly keyed connector. If necessary, add a diode in series with the microcontroller's power input to protect against polarity reversal.

Wiring 1

In this version, the microcontroller is ATmega328P with the same specifications, but with support for 480 pixels

RANGMANG ATmega 328 – 480 Pixels

' 30 inputs

' ADRGBSEM

```
' A00D00R000G000B000S000E000M000
' A01 TO A98  FOR ADDRESS MODULE  A99 FOR ALL
' IF D00 THEN SOLID COLOR
' R000 TO R255  RED COLOR
' G000 TO G255  GREEN COLOR
' B000 TO B255  BLUE COLOR
' S000 TO S480  RGB STRIP START POSITION FOR COLORED PIXEL
' E000 TO E480  RGB STRIP END POSITION FOR COLORED PIXEL
' M000 TO M098  M000 WITHOUT MEMORY SAVE  M001 SAVE SLOT 1
MEMORY

' IF D01 TO D38  DANCE
' D99  FOR RANDOM SELECT D01 TO D38 DANCE
' R000 TO R250  RED COLOR  R251 TO R255 FOR RANDOM RED COLOR
' G000 TO G250  GREEN COLOR  G251 TO G255 FOR RANDOM GREEN COLOR
' B000 TO B250  BLUE COLOR  B251 TO B255 FOR RANDOM BLUE COLOR
' S000 TO S999  RGB STRIP FIRST DELAY EFFECT (ms)
' E000 TO E999  RGB STRIP SECOND DELAY EFFECT (ms)
' M001 TO M480  SET HOW MANY PIXELS DANCING
RANGMANG ATmega128 – 999 Pixels
' 30 inputs
```

ADRGBSEM

```
' A00D00R000G000B000S000E000M000
' A01 TO A98  FOR ADDRESS MODULE  A99 FOR ALL
' IF D00 THEN SOLID COLOR
' R000 TO R255  RED COLOR
' G000 TO G255  GREEN COLOR
```

' B000 TO B255 BLUE COLOR

' S000 TO S480 RGB STRIP START POSITION FOR COLORED PIXEL

' E000 TO E480 RGB STRIP END POSITION FOR COLORED PIXEL

' M000 TO M098 M000 WITHOUT MEMORY SAVE M001 SAVE SLOT 1 MEMORY

' IF D01 TO D38 DANCE

' D99 FOR RANDOM SELECT D01 TO D38 DANCE

' R000 TO R250 RED COLOR R251 TO R255 FOR RANDOM RED COLOR

' G000 TO G250 GREEN COLOR G251 TO G255 FOR RANDOM GREEN COLOR

' B000 TO B250 BLUE COLOR B251 TO B255 FOR RANDOM BLUE COLOR

' S000 TO S999 RGB STRIP FIRST DELAY EFFECT (ms)

' E000 TO E999 RGB STRIP SECOND DELAY EFFECT (ms)

' M001 TO M999 SET HOW MANY PIXELS DANCING

Module Addressing Instructions

To address a module, simply follow the method below.

For example:

Address 2 → CH=02

Address 34 → CH=34

After sending the command, the module will respond and operate according to the specified address.

Please note:

Valid addresses range from 01 to 98.

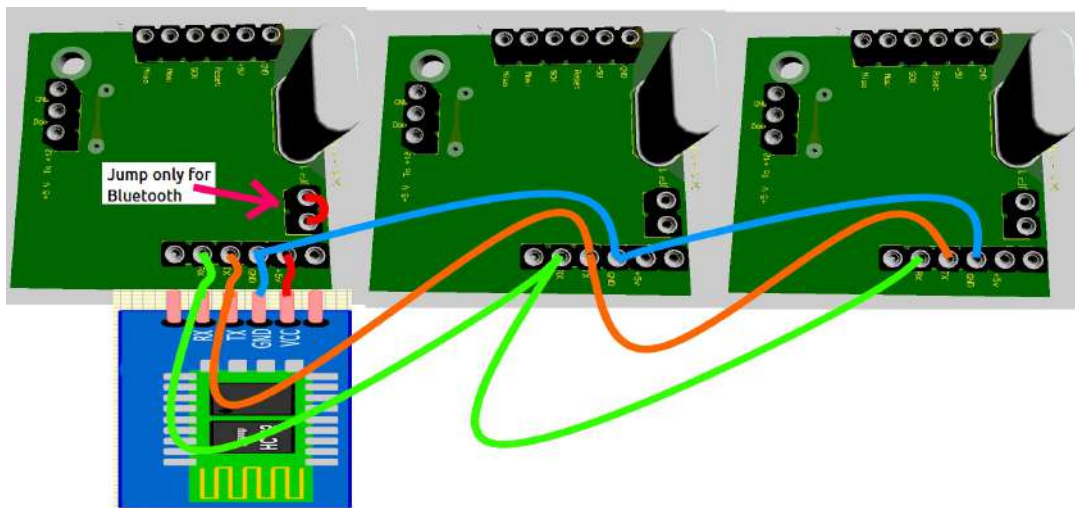
Address 99 is a shared address that applies to all modules.

Therefore, each module will respond and act to its assigned address (between 01 and 98) as well as to the shared address 99

The Rangmand209 modules can connect to each other for receiving and responding to serial data. It is sufficient to connect only one module to Bluetooth or a serial interface.

Do not connect the 5V pin of each module to the others, as this can lead to regulator errors, potential malfunctions, and system-wide failures. Only three pins—GND, RX, and TX—should be interconnected.

Pins	Connections
GND	GND1 → GND2 → GND3 → ... → GND98
RX	RX1 → RX2 → RX3 → ... → RX98
TX	TX1 → TX2 → TX3 → ... → TX98



Do not connect the 5V pin to prevent regulator errors and potential damage to the modules.

Keywords ws2811 ws2812
ws2812b soifdesign android-app blue

